The Mandelbrot Set

1 Definition

Fix a point $c = (a, b)$ of the plane, viewed as the complex number $c = a + ib$, and iterate the map

$$z_0 = 0, \qquad z_{n+1} = z_n^2 + c. \tag{1}$$

The *Mandelbrot set* M is the set of parameters c for which this sequence remains bounded. Writing $z_n = x_n + iy_n$ and separating real and imaginary parts gives the purely real recurrence used by the program:

$$x_0 = y_0 = 0, \qquad \begin{cases} x_{n+1} = x_n^2 - y_n^2 + a \\ y_{n+1} = 2x_n y_n + b. \end{cases} \tag{2}$$

2 The escape criterion

Membership asks about the sequence's behavior at infinity, which no finite computation can decide directly. Two facts rescue the programmer:

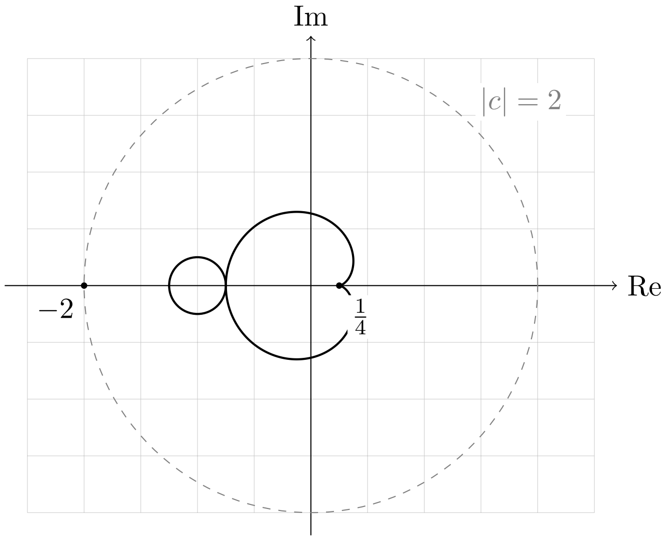
1. If $|z_n| > 2$ for some n, the sequence diverges. Indeed, once $|z| > 2$ and $|z| \geq |c|$,

   $$|z^2 + c| \;\geq\; |z|^2 - |c| \;\geq\; |z|^2 - |z| \;=\; |z|\,(|z| - 1) \;>\; |z|,$$

   and the modulus grows at an accelerating rate thereafter. Consequently M is contained in the closed disk of radius 2, and the program's test $x_n^2 + y_n^2 > 4$ is a sound divergence certificate.

2. In practice, points outside M tend to escape quickly, so testing the first k terms (the book uses $k = 100$) classifies all but a thin sliver near the boundary correctly.

The *escape time* of a point $c \notin M$ is the first index n with $|z_n| > 2$. It measures how close c is to the set — distant points escape in a handful of steps, points hugging the boundary may take hundreds — and coloring pixels by escape time produces the familiar glowing halos around the fractal.



The two largest pieces of M: the main cardioid and the period-2 disk, inside the escape disk $|c| \leq 2$.

3 Properties

- **A cardioid again.** The big heart-shaped body of M is an exact cardioid: the parameters $c = \frac{e^{i\theta}}{2} - \frac{e^{2i\theta}}{4}$ for which the iteration has an attracting fixed point. Our previous quiz drew its cousin.
- **Bounded but intricate.** M fits in the disk of radius 2, is connected (Douady–Hubbard, 1982), and its boundary is so wrinkled that its Hausdorff dimension is 2 (Shishikura, 1998).
- **On the real line.** $M \cap \mathbb{R} = [-2, \frac{1}{4}]$: the program's test points $(0,0)$, $(-1,0)$, $(-2,0)$ are members, while $(1,0)$ escapes in three steps.
- **Self-similarity.** Zooming anywhere near the boundary reveals miniature copies of the whole set — the hallmark of a fractal.

4 From mathematics to pixels

The program scans every pixel (w, h) of the window, maps it affinely to a point $(a, b) \in [-2, 2]^2$, and iterates the recurrence at most k times. If $x_n^2 + y_n^2$ never exceeds 4, the pixel is considered a member and plotted. The precision of the picture depends only on k: raising it sharpens the boundary at the price of more arithmetic on the points that never escape.